Arab Academy for Science, Technology & Maritime Transport

College of Artificial Intelligence ———— Alamein Campus.

Course	DS414	Information Retrieval and Search Engines
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1. Literature Review

1.1. Multi-Agent Architectures

- Arxiv 2304.08085 (2023): Proposed a multi-agent coordination framework where specialized agents collaborate to solve complex tasks, demonstrating the value of modular decomposition.
- DARD (Amazon Science, 2023): Introduced a distributed multi-agent dialogue architecture that aggregates responses from multiple agents for improved robustness and task handling.

1.2. Multimodal Fusion Models

- Arxiv 2311.17136 (2023): Developed a multimodal LLM integrating vision and text via early fusion with cross-modal attention, enabling deep modality interaction.
- Arxiv 2410.05160 (2024): Proposed hierarchical multimodal fusion that combines features at different abstraction levels, showing superior performance on complex reasoning tasks.

1.3. Retrieval-Augmented Systems

- Arxiv 2511.12586 (2025): Introduced a retrieval-augmented multimodal generation framework using late fusion to ground outputs in external knowledge for factual accuracy.
- Arxiv 2505.02024 (2025): Enhanced multimodal retrieval with adaptive cross-modal attention, allowing dynamic focus on relevant query parts for improved results.

1.4. Planning & Reasoning Agents

- Arxiv 2503.19868 (2025): Proposed autonomous planning agents capable of using external tools (APIs, calculators) to execute multi-step tasks beyond simple Q&A.
- Arxiv 2511.08181 (2025): Integrated memory and reasoning modules for long-horizon tasks, enabling agents to maintain context over extended interactions.

1.5. Evaluation & Benchmarking

- Arxiv 2404.07972 (OSWorld, 2024): Created a benchmark for multimodal agents on real computer tasks, revealing a massive gap between human (72.36%) and AI (12.24%) performance.
- ACL EvalMG (LLaVA-RE, 2025): Proposed a standardized evaluation protocol for binary image-text relevancy using MLLMs with task instructions and in-context learning.

2. Research Questions with reasons (Why this research question)

- ***RQ1 (Architecture & Fusion): How can a modular agent be designed to process, fuse, and reason over text, image, and audio for task-oriented IR?***
    - *Rationale:* Literature shows advanced components (coordination, fusion, planning) but no cohesive framework integrating all three for actionable task completion.
- ***RQ2 (Task Modeling): How can task completion be modeled as decision-making over fused multimodal representations?***
    - *Rationale:* Existing planning agents (Arxiv 2503.19868) are text-only; this extends planning to agents whose world understanding comes from fused multimodal inputs.
- ***RQ3 (Retrieval Integration): How can retrieval-augmented generation be integrated with multimodal fusion and task planning for knowledge-grounded actions?***
    - *Rationale:* RAG frameworks (Arxiv 2511.12586) lack integration with task planners; this combines factual grounding with goal-oriented execution.
- ***RQ4 (Evaluation): What comprehensive framework can assess both retrieval effectiveness and task completion performance holistically?***
    - *Rationale:* OSWorld provides task benchmarking; LLaVA-RE provides relevance metrics. No unified framework combines both aspects.

3. Hypotheses with reasons (Why is this hypothesis)

- ***H1 (Fusion Superiority): Hierarchical or cross-modal attention fusion will outperform concatenation or late fusion on complex reasoning tasks.***
    - *Rationale:* Based on Arxiv 2410.05160 and Arxiv 2311.17136, sophisticated fusion captures nuanced modality interactions better than simple methods.
- ***H2 (RAG Integration): Integrating a retrieval-augmented generation module will improve task completion rates for knowledge-intensive tasks.***
    - *Rationale:* Derived from Arxiv 2511.12586, external knowledge grounding provides critical context missing from parametric memory alone.
- ***H3 (Tool Augmentation): A tool-augmented reasoning pipeline will achieve higher success rates on multi-step tasks than internal processing alone.***

    - *Rationale:* Informed by Arxiv 2503.19868, offloading sub-tasks to dedicated tools is more efficient and accurate than forcing the core model to handle everything.
- ***H4 (Multi-Metric Evaluation): Combining task success rate, completion time, and cross-modal relevance will provide more valid assessment than success rate alone.***
    - *Rationale:* OSWorld and LLaVA-RE show unidimensional metrics mask weaknesses; composite scores better reflect real-world usefulness.